

Question #1: Consider the length- L moving-average FIR system (Hint: use lecture 12)

$$h[n] = \frac{1}{L} \sum_{k=0}^{L-1} \delta[n-k] \quad (\text{take } L = 5).$$

(a) Derive the frequency response $H(e^{j\omega})$ and show it can be written as

$$H(e^{j\omega}) = \frac{\sin\left(\frac{L\omega}{2}\right)}{L \sin\left(\frac{\omega}{2}\right)} e^{-j\omega \frac{L-1}{2}}.$$

$$H(e^{j\omega}) = e^{j\omega}$$

$$\begin{aligned} \sum_{k=0}^{L-1} r^k &= \frac{1-r^L}{1-r} = \frac{1}{L} * \frac{1-e^{-j\omega L}}{1-e^{-j\omega}} = \frac{1}{L} \frac{e^{-j\omega \frac{L-1}{2}} \left(e^{j\omega \frac{L-1}{2}} - e^{-j\omega \frac{L-1}{2}} \right)}{e^{-j\omega \frac{L-1}{2}} \left(e^{j\omega \frac{L-1}{2}} - e^{-j\omega \frac{L-1}{2}} \right)} \\ &= \frac{\frac{1}{L} \left(e^{-j\omega \frac{L-1}{2}} 2j \sin\left(\frac{\omega L}{2}\right) \right)}{e^{-j\omega \frac{L-1}{2}} 2j \sin\left(\frac{\omega}{2}\right)} = \frac{1}{L} e^{j\omega \left(\frac{L-1}{2}\right)} * \frac{\sin\left(\frac{\omega L}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} = \frac{e^{-j\omega \frac{L-1}{2}}}{2} * \frac{1 * \sin\left(\frac{\omega L}{2}\right)}{L * \sin\left(\frac{\omega}{2}\right)} \end{aligned}$$

Identify the magnitude $|H(e^{j\omega})|$ and phase $\angle H(e^{j\omega})$.

$$1 * \left| \frac{\sin\left(\frac{\omega L}{2}\right)}{L \sin\left(\frac{\omega}{2}\right)} \right| = \left| \frac{\sin\left(\frac{5\omega}{2}\right)}{5 \sin\left(\frac{\omega}{2}\right)} \right|$$

(b) For $L = 5$, list all ω in $[-\pi, \pi]$ where $|H(e^{j\omega})| = 0$. Classify the filter as low-, high-, or band-pass and justify briefly from $|H|$.

$$\begin{aligned} H(e^{j\omega}) &= \left\{ -\frac{\omega(L-1)}{2} \rightarrow |\omega| \leq \frac{2\pi}{3}, |\omega| > \frac{2\pi}{3} \right. \\ &= \left\{ -\frac{\omega(L-1)}{2} + \pi \text{ when } \frac{\sin\left(\frac{5\omega}{2}\right)}{5 \sin\left(\frac{\omega}{2}\right)} < 0 \right. \end{aligned} \quad \frac{2\pi}{5} < |\omega| < \frac{4\pi}{5}$$

(c) If the input is $x[n] = A \cos(\omega_0 n + \phi)$, find the output $y[n]$ in the form

$$y[n] = A |H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0})).$$

Evaluate the amplitude and phase shift numerically for $\omega_0 = \frac{\pi}{3}$.

$$\begin{aligned} |H(e^{j\omega})| &= \left| \frac{\sin\left(\frac{5 * \frac{\pi}{3}}{2}\right)}{5 \sin\left(\frac{\frac{\pi}{3}}{2}\right)} \right| = \left| \frac{\sin\left(\frac{5\pi}{6}\right)}{5 \sin\left(\frac{\pi}{6}\right)} \right| = \frac{\frac{1}{2}}{5 * \frac{1}{2}} = \frac{1}{5} \\ &= -2 * \frac{\pi}{3} = -\frac{2\pi}{3} \end{aligned}$$

$$y[n] = A * \frac{1}{5} \cos\left(w_0 n t \phi - \frac{2\pi}{3}\right)$$

Question #2: Two LTI systems are cascaded: a moving average and a first-difference,

$$h_1[n] = \frac{1}{3}(\delta[n] + \delta[n-1] + \delta[n-2]), \quad h_2[n] = \delta[n] - \delta[n-1]$$

- (a) Compute the overall impulse response $h_T[n] = h_1[n] * h_2[n]$ explicitly. Then find

$$H_T(e^{j\omega}) = H_1(e^{j\omega})H_2(e^{j\omega})$$

and simplify its magnitude expression using $|1 - e^{-j\omega}| = 2 \left| \sin \frac{\omega}{2} \right|$.

- (b) With input $x[n] = \cos(\frac{\pi}{3}n)$, determine the steady-state output amplitude and phase (use Problem 1's sinusoid-through-LTI idea).
- (c) Is the overall system causal? Is it FIR? Briefly justify. Comment qualitatively on whether H_T behaves more like a differentiator (high-pass) at low ω .

Question #3: Let $x[n] = \cos(\frac{\pi}{4}n) w[n]$, where $w[n]$ is a length- $N = 16$ rectangular window ($w[n] = 1$ for $0 \leq n \leq 15$, else 0).

- (a) Write the DTFT definition $X(e^{j\omega})$ for a general $x[n]$ and state its 2π -periodicity. Then explain in one or two sentences how the DFT samples the DTFT at $\omega_k = 2\pi k/N$.
- (b) For this $x[n]$, specify the DFT bin locations ω_k and identify the nearest bin(s) to $\omega_0 = \frac{\pi}{4}$. Will there be spectral leakage? Why?
- (c) Suppose we instead choose $N = 8$. Compare how the bin spacing changes and how that affects leakage for the same tone.
- (d) **MATLAB (short):** Write code to compute $X = \text{fft}(x, N)$ for $N = 16$, plot $|X[k]|$, and annotate the bin nearest to ω_0 .

Question #4: Let $x[n] = [1, 2, 1, 0]$ and $h[n] = [1, -1, 1]$.

- (a) What is the length of the *linear* convolution $y[n] = x[n] * h[n]$? Compute it by hand or by a structured table.
- (b) Show how to obtain the same linear convolution using DFTs: choose the smallest N so that circular convolution equals linear convolution, then outline the zero-padding and FFT/IFFT steps.
- (c) Now take $N = 4$ instead and compute the *circular* convolution (by hand or using the DFT idea). Identify the indices where time-aliasing occurs relative to the linear result.
- (d) **MATLAB (short):** Provide code that (i) computes $\text{conv}(x, h)$, (ii) computes circular convolution with $\text{ifft}(\text{fft}(x, N) * \text{fft}(h, N))$ for $N = 4$ and for the zero-padded N from part (b), and (iii) compares the results.